



APPLICATION OF NEURAL NETWORK BASED OFFLINE LEARNING FOR CONTROL OF A QUADROTOR HELICOPTER

Project Report for ENEL 667 Intelligent Control

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1 INTRODUCTION

The purpose of this project is to derive a control using the intelligent control approach with offline training of nonlinear system model functions. The control is then applied in a closed loop simulation.

The system I chose to model is a quadrotor helicopter (quadrotor). Conventional helicopters normally have two rotors: the larger of the two provides lift and a smaller one located laterally counteracts the torque of the larger one. The larger rotor design is complex, owing to its ability to provide directional motion. Quadrotors offer benefits over conventional helicopters in their simplicity of design. Quadrotors have four rotors, equally spaced outboard from the main body. Unlike conventional helicopters, quadrotor rotors do not have variable pitch blades. This greatly simplifies the mechanical aspect of the physical design.

Control of quadrotors primarily consists of four inputs, namely the speed for each rotor, and these ultimately provide six degrees of freedom of motion (three translational and three rotational). The dynamics involved are complicated and nonlinear owing to aerodynamic and gyroscopic effects. Also, there is little friction to dampen motions. These factors make control of quadrotors challenging.

The scope of this project is limited to the stability of the quadrotor in a hovering mode, as well as its stability and tracking ability when subjected to a trajectory input. In this case the chosen trajectory is an oscillation (sinusoid) of the quadrotor pitch.

2 MODEL

2.1 MATHEMATICAL MODEL

Figure 1 illustrates the Cartesian coordinate system (body frame) and sign convention for quadrotor rotations and translation as well as rotor spin directions.

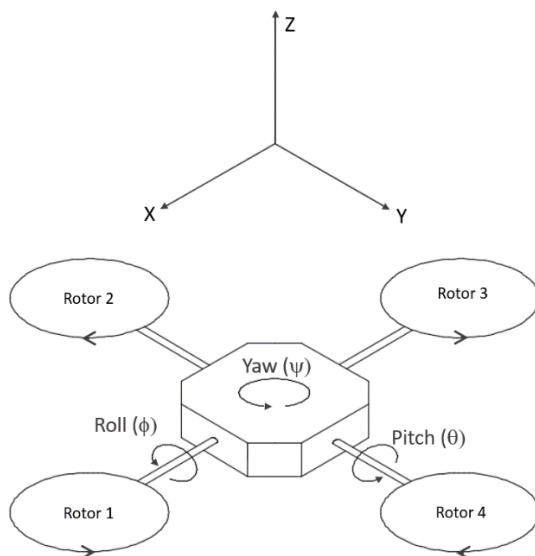


Figure 1. Quadrotor coordinate system and sign convention.

For this project simplified dynamics equations were used [1] [2]. Translation in the x and y directions was considered outside the scope of the project; however, motion in the z direction was included since adjusting rotor speeds directly impacts the elevation. A representative state space model of the system dynamics includes eight states: $z, \phi, \theta, \psi, \dot{z}, \dot{\phi}, \dot{\theta},$ and $\dot{\psi}$.

$$\begin{aligned}
 \ddot{z} &= -g \cos\phi \cos\theta + \frac{1}{m} F_z \\
 \ddot{\phi} &= \dot{\theta}\dot{\psi} \frac{J_y - J_z}{J_x} - \frac{J_r}{J_x} \dot{\theta}\Omega_r + \frac{1}{J_x} T_\phi \\
 \ddot{\theta} &= \dot{\phi}\dot{\psi} \frac{J_z - J_x}{J_y} + \frac{J_r}{J_y} \dot{\phi}\Omega_r + \frac{1}{J_y} T_\theta \\
 \ddot{\psi} &= \dot{\phi}\dot{\theta} \frac{J_x - J_y}{J_z} + \frac{1}{J_z} T_\psi
 \end{aligned} \tag{1}$$

Where,

z = elevation, m

$\dot{z}$ = velocity in z direction, m/s

$\ddot{z}$ = acceleration in z direction, m/s²

ϕ = roll angle, radians

$\dot{\phi}$ = roll rate, rad/s

$\ddot{\phi}$ = roll angular acceleration, rad/s²

θ = pitch angle, radians

$\dot{\theta}$ = pitch rate, rad/s

$\ddot{\theta}$ = pitch angular acceleration, rad/s²

ψ = yaw angle, radians

$\dot{\psi}$ = yaw rate, rad/s

$\ddot{\psi}$ = yaw angular acceleration, rad/s²

J_x = quadrotor moment of inertia in x direction, kgm²

J_y = quadrotor moment of inertia in y direction, kgm²

J_z = quadrotor moment of inertia in z direction, kgm²

J_r = single rotor moment of inertia, kgm²

Ω_r = sum of rotor speeds, rad/s

g = gravitation constant, m/s²

m = mass, kg

F_z = force in z direction, N

T_ϕ = torque about x axis, Nm

T_θ = torque about y axis, Nm

T_ψ = torque about z axis, Nm

In the first of Eq. 1 the z direction was fixed to the quadrotor body frame rather than the earth frame. Constants were used for the inertia terms to simplify formulas in the computer code:

$$A_1 = (J_y - J_z)/J_x, a_2 = -J_r/J_x, a_3 = (J_z - J_x)/J_y, a_4 = J_r/J_y, a_5 = (J_x - J_y)/J_z$$

Control inputs for this project are chosen to be F_z , T_ϕ , T_θ , and T_ψ , rather than voltages to the motors, in order to limit the scope of the project; however, rotor speeds are required for the dynamics equations and are therefore included in the performance results.

Rotor speeds are related to F_z and torques as follows.

$$\begin{aligned} F_z &= b (\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2) \\ T_\phi &= L b (\Omega_4^2 - \Omega_2^2) \\ T_\theta &= L b (\Omega_3^2 - \Omega_1^2) \\ T_\psi &= d (\Omega_1^2 + \Omega_3^2 - \Omega_2^2 - \Omega_4^2) \end{aligned} \quad (2)$$

Where,

$\Omega_1, \Omega_2, \Omega_3, \Omega_4$ = rotor speeds, rad/s

b = thrust factor, Ns^2

L = distance from quadrotor center to rotor, m

d = yaw rate drag factor, Nms^2

Solving Eq. 2 for rotor speeds yields

$$\begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix} = \begin{bmatrix} 0.25 & 0 & -0.5 & 0.25 \\ 0.25 & -0.5 & 0 & -0.25 \\ 0.25 & 0 & 0.5 & 0.25 \\ 0.25 & 0.5 & 0 & -0.25 \end{bmatrix} \begin{bmatrix} F_z \\ T_\phi \\ T_\theta \\ T_\psi \end{bmatrix} \quad (3)$$

Since the state space system includes eight states, auxiliary state stability equations were employed.

$$\begin{aligned} x_1 &= \Lambda z + \dot{z} \\ x_2 &= \Lambda \phi + \dot{\phi} \\ x_3 &= \Lambda \theta + \dot{\theta} \\ x_4 &= \Lambda \psi + \dot{\psi} \end{aligned} \quad (4)$$

Where,

$\Lambda = \text{a constant (1), } 1/s$

Regulation of these auxiliary states results in stability in each of the original states. Equating the derivatives of the auxiliary states to the dynamics equations (Eq. 1) results in the following:

$$\begin{aligned}
 \dot{x}_1 &= \Lambda \dot{z} + \ddot{z} = \Lambda \dot{z} - g \cos \phi \cos \theta + \frac{1}{m} F_z \\
 \dot{x}_2 &= \Lambda \dot{\phi} + \ddot{\phi} = \Lambda \dot{\phi} + \dot{\theta} \dot{\psi} a_1 + \dot{\theta} \Omega_r a_2 + \frac{1}{J_x} T_\phi \\
 \dot{x}_3 &= \Lambda \dot{\theta} + \ddot{\theta} = \Lambda \dot{\theta} + \dot{\phi} \dot{\psi} a_3 + \dot{\phi} \Omega_r a_4 + \frac{1}{J_y} T_\theta \\
 \dot{x}_4 &= \Lambda \dot{\psi} + \ddot{\psi} = \Lambda \dot{\psi} + \dot{\phi} \dot{\theta} a_5 + \frac{1}{J_z} T_\psi
 \end{aligned} \tag{5}$$

A modification was made the third equation in Eq. 5 to allow tracking of a trajectory. The auxiliary state was replaced with an auxiliary state error relating actual to desired terms.

$$\begin{aligned}
 z_3 &= x_3 - x_{3 \text{ des}} \\
 \dot{z}_3 &= (\Lambda \dot{\theta} + \ddot{\theta}) - (\Lambda \dot{\theta}_{des} + \ddot{\theta}_{des}) \\
 \dot{z}_3 &= \Lambda (\dot{\theta} - \dot{\theta}_{des}) + (\ddot{\theta} - \ddot{\theta}_{des}) \\
 \dot{\tilde{x}}_3 &= \dot{z}_3 = \Lambda (\dot{\theta} - \dot{\theta}_{des}) + \dot{\phi} \dot{\psi} \frac{J_z - J_x}{J_y} + \frac{J_r}{J_y} \dot{\phi} \Omega_r - \ddot{\theta}_{des} + \frac{1}{J_y} T_\theta
 \end{aligned} \tag{6}$$

The resulting state space system is

$$\begin{aligned}
 \dot{x} &= f(x) + Bu \\
 \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{\tilde{x}}_3 \\ \dot{x}_4 \end{bmatrix} &= \begin{bmatrix} \Lambda \dot{z} - g \cos \phi \cos \theta \\ \Lambda \dot{\phi} + \dot{\theta} \dot{\psi} a_1 + \dot{\theta} \Omega_r a_2 \\ \Lambda (\dot{\theta} - \dot{\theta}_{des}) + \dot{\phi} \dot{\psi} a_3 + \dot{\phi} \Omega_r a_4 - \ddot{\theta}_{des} \\ \Lambda \dot{\psi} + \dot{\phi} \dot{\theta} a_5 \end{bmatrix} + \begin{bmatrix} 1/m & 0 & 0 & 0 \\ 0 & 1/J_x & 0 & 0 \\ 0 & 0 & 1/J_y & 0 \\ 0 & 0 & 0 & 1/J_z \end{bmatrix} \begin{bmatrix} F_z \\ T_\phi \\ T_\theta \\ T_\psi \end{bmatrix}
 \end{aligned} \tag{7}$$

2.2 STABILITY ANALYSIS

A suitable Lyapunov function candidate for the state space system in Eq. 7 is

$$V = \frac{1}{2} x^T B^{-1} x \quad (8)$$

The derivative of this function is then

$$\begin{aligned} \dot{V} &= x^T B^{-1} \dot{x} \\ \dot{V} &= x^T B^{-1} (f(x) + Bu) \\ \dot{V} &= x^T (B^{-1} f(x) + u) \end{aligned} \quad (9)$$

Approximation of the nonlinear term $B^{-1}f(x)$ was accomplished using neural network function approximation algorithms: Cerebellar Model Articulation Controller (CMAC) and Multi-layer Perceptron (MLP).

$$B^{-1}f(x) = \vartheta(x) w + d(x) \quad (10)$$

In Eq. 10 the $d(x)$ term represents the error between the actual function value and the approximation.

A suitable control signal candidate that eliminates the nonlinear term and adds feedback is

$$u = -\vartheta(x) w - K x \quad (11)$$

Where,

K = gains matrix (diagonal)

Adding the control signal u to the third equation in Eq. 9 results in

$$\begin{aligned} \dot{V} &= x^T (\vartheta(x) w + d(x) - \vartheta(x) w - k x) \\ \dot{V} &= x^T (d(x) - k x) \end{aligned} \quad (12)$$

In a worst case scenario, and taking norms, $\dot{V}$ becomes

$$\dot{V} = \|x\|(d_{max} - K_{min} \|x\|) \quad (13)$$

$\|x\|$ is uniformly ultimately bounded (UUB), with $\dot{V} < 0$ when

$$\|x\| > \frac{\|d_{max}\|}{K_{min}} \quad (14)$$

$\|d_{max}\|$ represents the norm of the maximum estimation errors for each of the four function approximations, and K_{min} is the minimum eigenvalue in the diagonal gains matrix K , or equivalently the minimum gain value.

2.3 STATIC TRAINING

There are four nonlinear terms that require approximation. Recall this approximation is of $B^{-1}f(x)$ in Eq. 9. The four functions are explicitly

$$\begin{bmatrix} f_1(x) \\ f_2(x) \\ f_3(x) \\ f_4(x) \end{bmatrix} = B^{-1}f(x) = \begin{bmatrix} m(\Lambda\dot{z} - g \cos\phi \cos\theta) \\ J_x(\Lambda\dot{\phi} + \dot{\theta}\psi a_1 + \dot{\theta}\Omega_r a_2) \\ J_y(\Lambda(\dot{\theta} - \dot{\theta}_{des}) + \dot{\phi}\psi a_3 + \dot{\phi}\Omega_r a_4 - \ddot{\theta}_{des}) \\ J_z(\Lambda\dot{\psi} + \dot{\phi}\dot{\theta} a_5) \end{bmatrix} \quad (15)$$

The variables in the functions in Eq. (15) are $\dot{z}$, ϕ , and θ for function 1, $\dot{\phi}$, $\dot{\theta}$, $\dot{\psi}$, and Ω_r for function 2, $\dot{\theta}$, $\dot{\theta}_{des}$, $\dot{\phi}$, $\dot{\psi}$, Ω_r , and $\ddot{\theta}_{des}$ for function 3, and $\dot{\psi}$, $\dot{\phi}$, and $\dot{\theta}$ for function 4. Suitable ranges for these variables were established for the purpose of stability control and following predefined trajectories in the control simulation work. Initially MLP was tested on all four functions (RHS of Eq. (15)), with the following results:

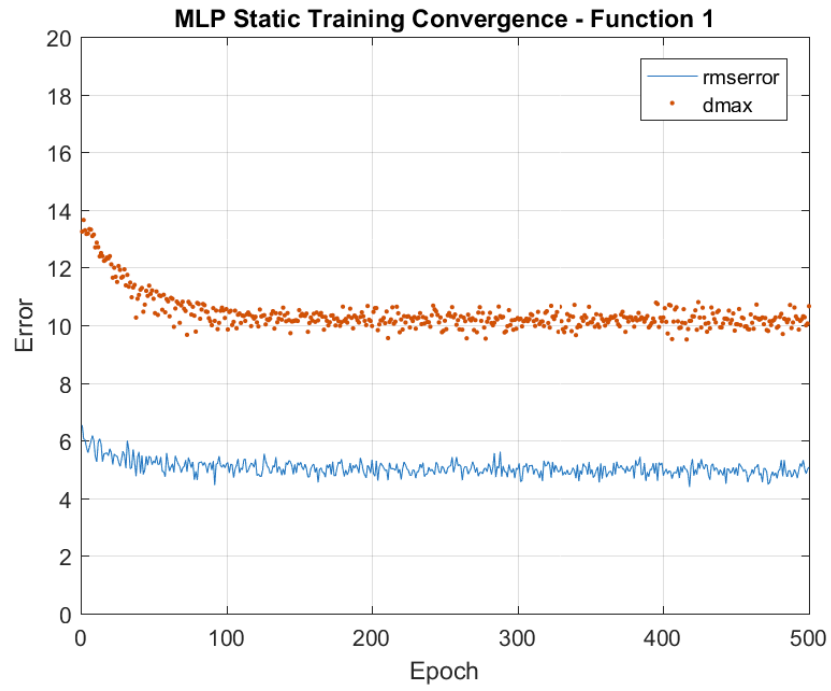


Figure 2. Static MLP training for function 1.

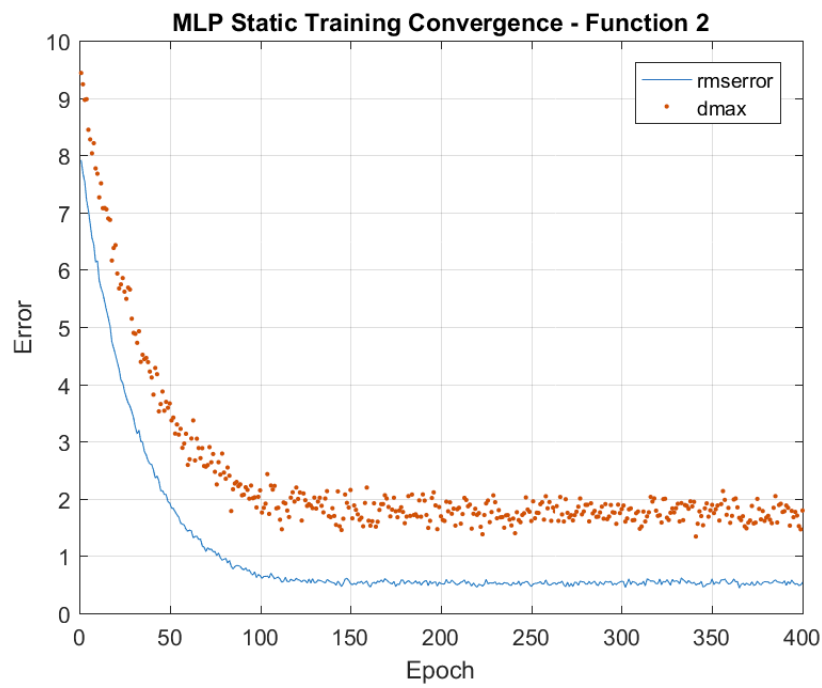


Figure 3. Static MLP training for function 2.

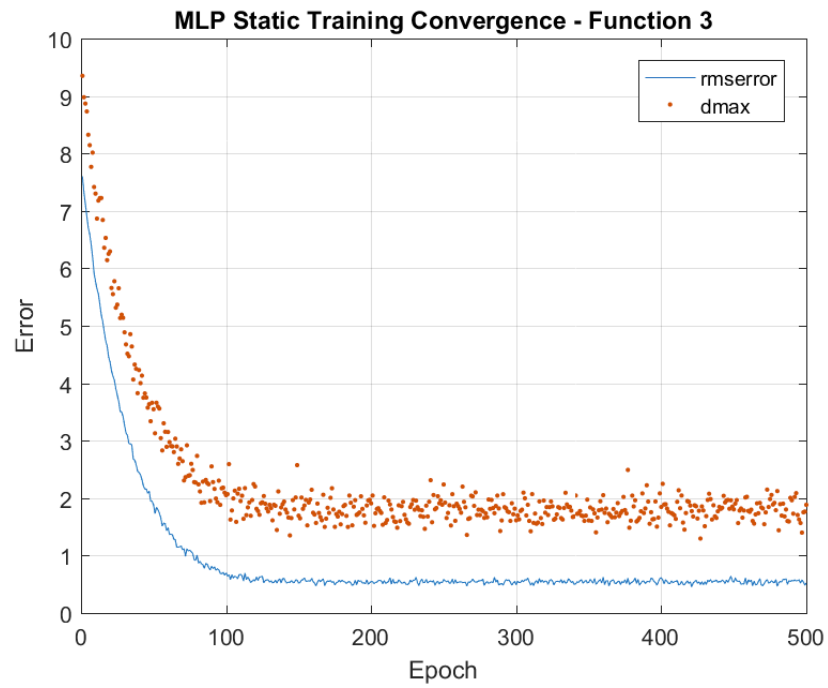


Figure 4. Static MLP training for function 3.

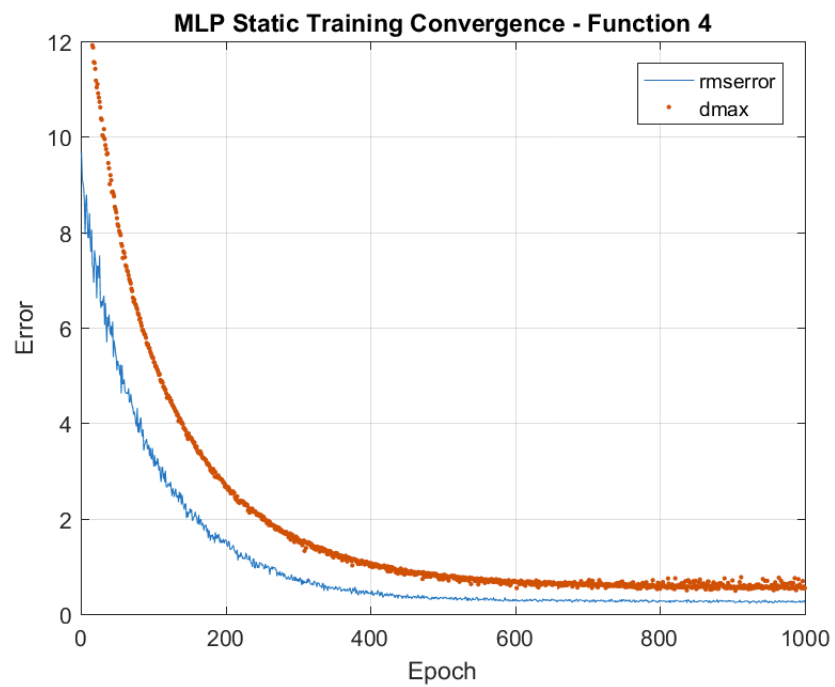


Figure 5. Static MLP training for function 4.

The approximation of function 1 using MLP was considered unsatisfactory. CMAC was then employed with better results

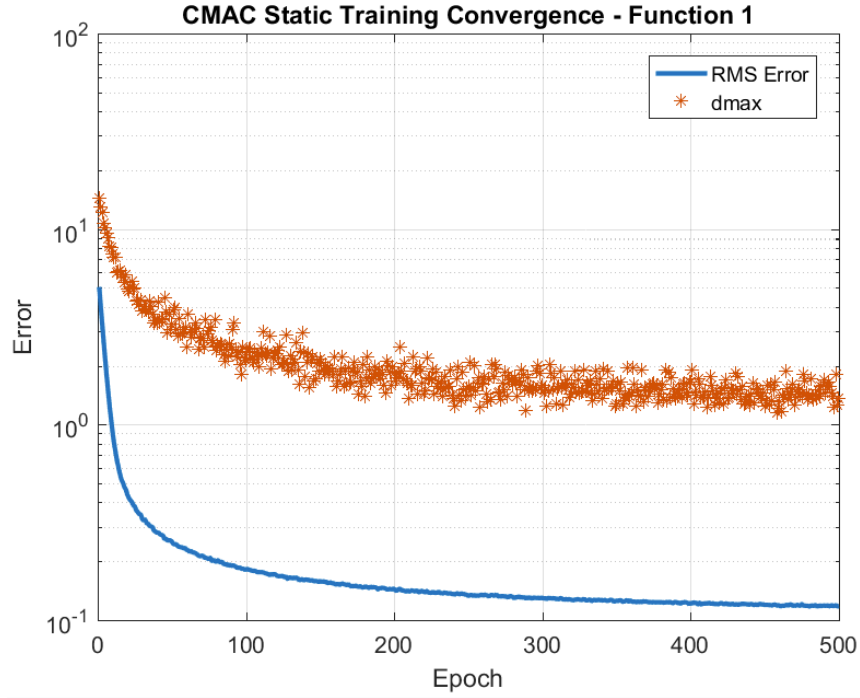


Figure 6. Static CMAC training for function 1.

Returning to the stability analysis of Section 2.2, the static training results facilitate solution of Eq. 14

$$\|x\| > \frac{\|d_{max}\|}{K_{min}} \cong \frac{3}{2} \quad (16)$$

Ongoing work includes testing CMAC function approximation performance on functions 2, 3, and 4. The mixed CMAC and MLP function approximation work done to date yields satisfactory performance as will be shown below.

2.4 CONTROL SIMULATION

Simulation modelling was done primarily in Simulink (see diagram in Appendix I). General generically established quadrotor parameters used in the simulations are shown in Table 1.

Table 1. Quadrotor physical parameters.

Parameter	Symbol	Value	Units
Quadrotor moment of inertia in x direction	Jx	0.01	Kg m ²
Quadrotor moment of inertia in y direction	Jy	0.01	Kg m ²
Quadrotor moment of inertia in z direction	Jz	0.02	Kg m ²
Single rotor moment of inertia	Jr	0.0001	Kg m ²
Thrust factor	b	6.13E-05	N s ²
Distance from quadrotor center to rotor	L	0.25	m
Yaw rate drag factor	d	1.00E-06	N m s ²
Quadrotor mass	m	1	kg

A summary of the closed loop simulation parameters are shown in Table 2, with performance plots shown below the table.

Table 2. Quadrotor simulation parameters.

Parameter	Units	Simulation Cases		
		1	2	3
<i>Initial conditions</i>				
Initial elevation	m	1	1	1
Velocity in z direction	m/s	0	0	0
Roll angle	rad	1	1	1
Roll rate	rad/s	0	0	0
Pitch angle	rad	-1	-1	-1
Pitch rate	rad/s	0	0	0
Yaw angle	rad	1	1	1
Yaw rate	rad/s	0	0	0
<i>Feedback gains</i>				
K ₁		3	3	3
K ₂		2	2	2
K ₃		2	2	2
K ₄		2	2	2
<i>Sinusoid pitch trajectory</i>				
Amplitude	rad	na	1	1.5
Frequency	rad/s	na	$\pi/2$	2π

2.4.1 Simulation Case 1

The first closed loop simulation case was without a trajectory to follow in order to test appropriate gains and general stability given initial conditions.

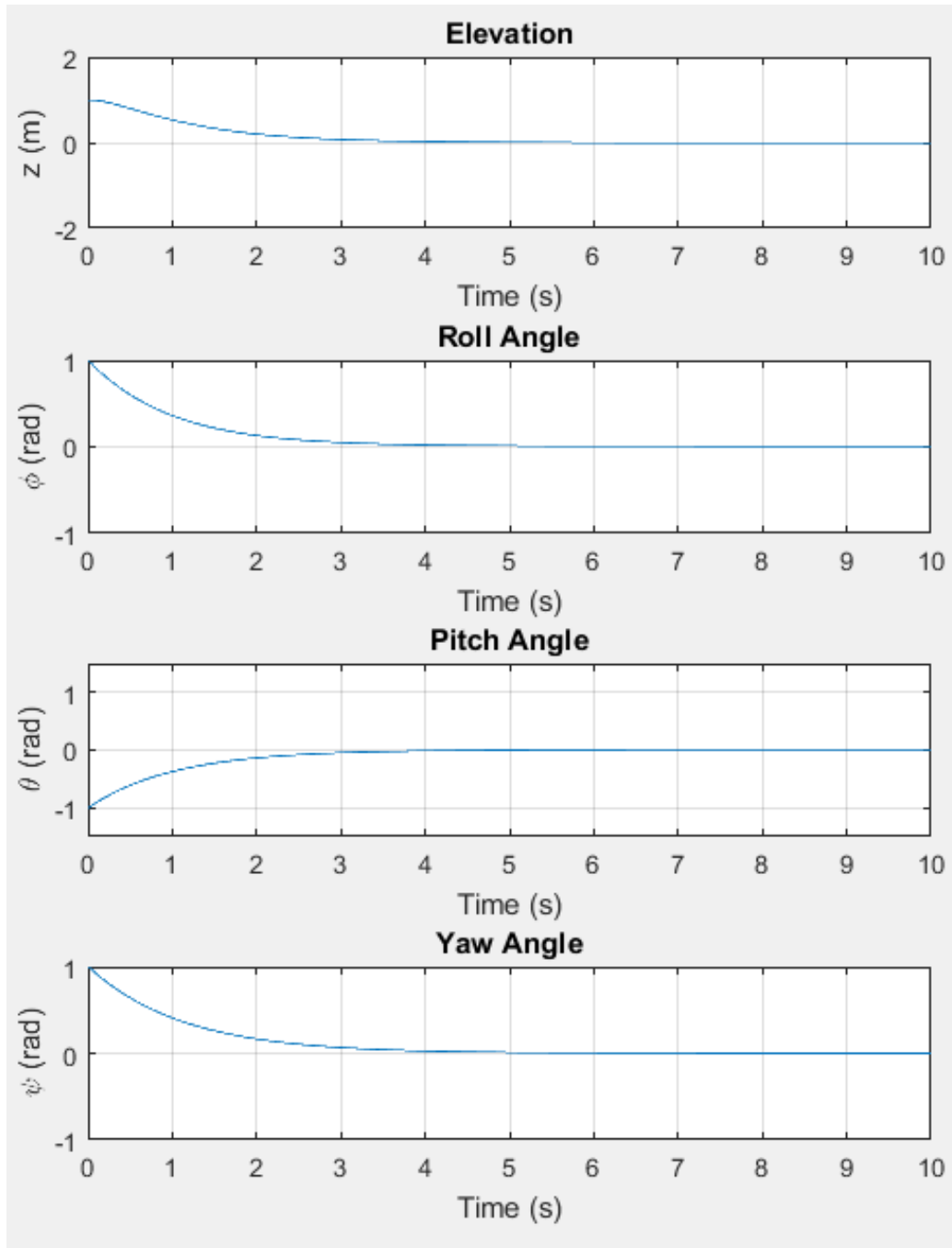


Figure 7. Closed loop simulation case 1; elevation and angles.

Settling time for an initial angle of one radian (57°) in all three axes is 3 to 4 seconds.

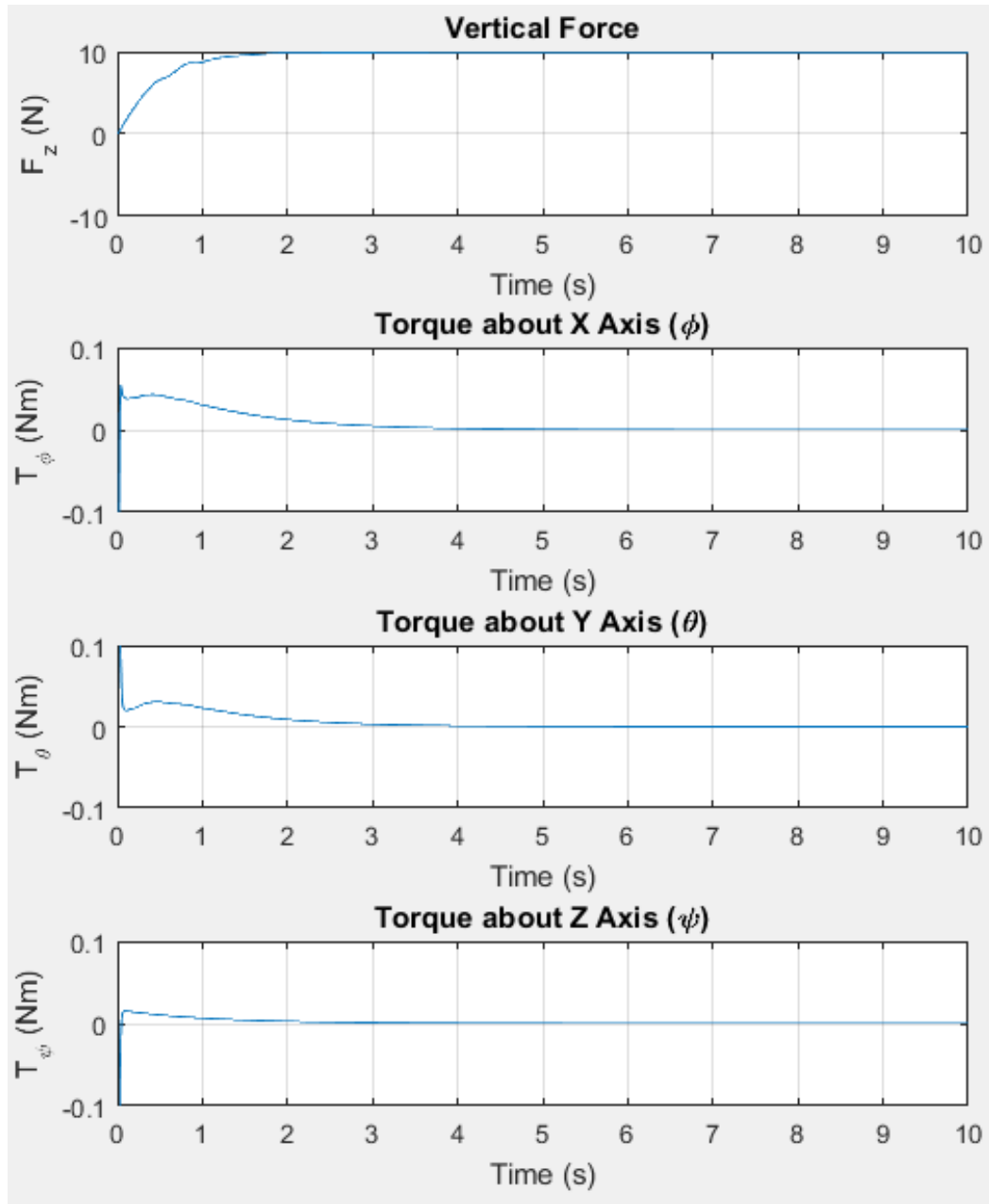


Figure 8. Closed loop simulation case 1; input vertical force and torques.

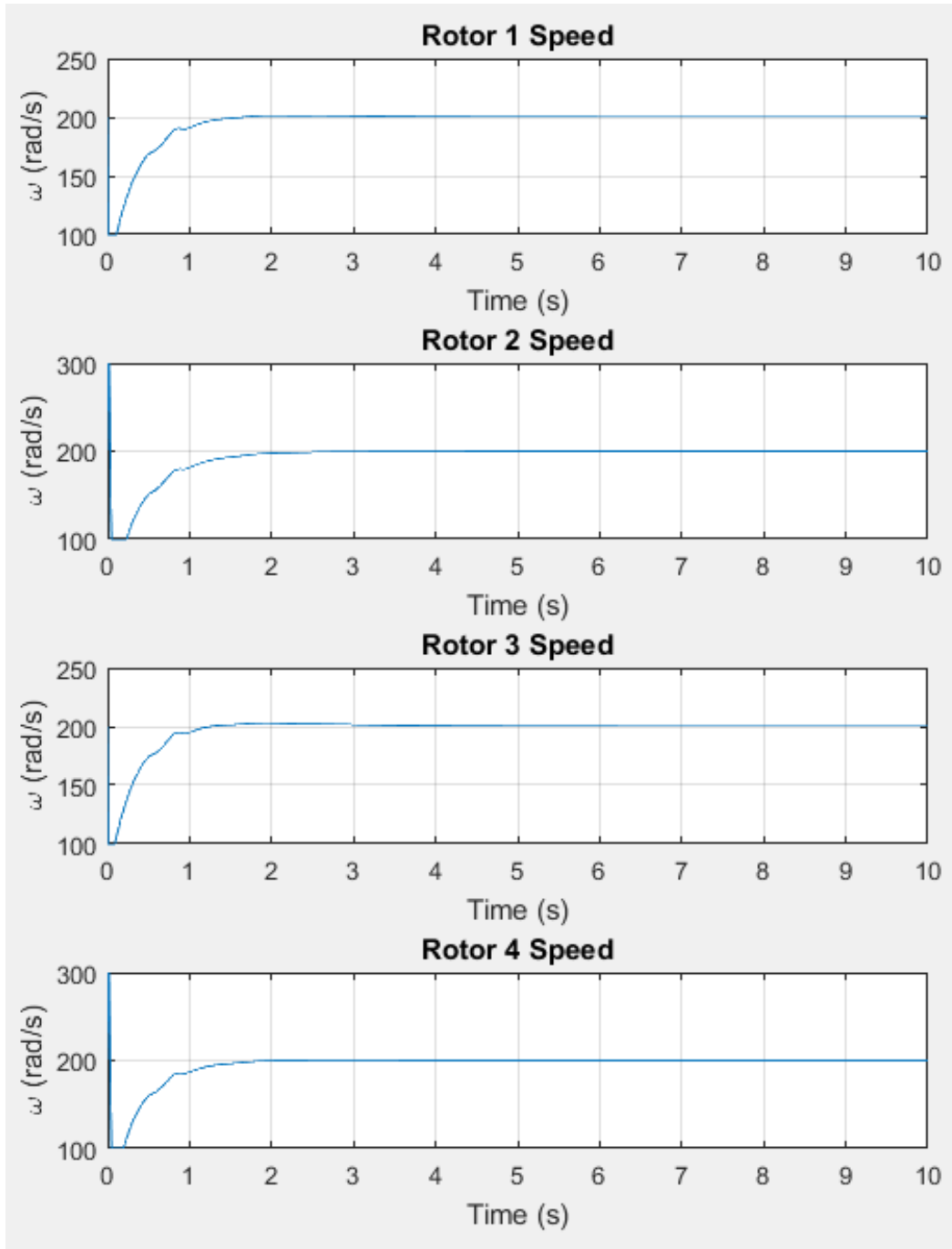


Figure 9. Closed loop simulation case 1; rotor speeds.

Rotor speeds are initially in a low state due to the initial condition of 1 m elevation.

2.4.2 Simulation Case 2

The next closed loop simulation was performed to show the quadrotor's ability to follow the first pitch angle sinusoidal trajectory input as detailed in Table 2.

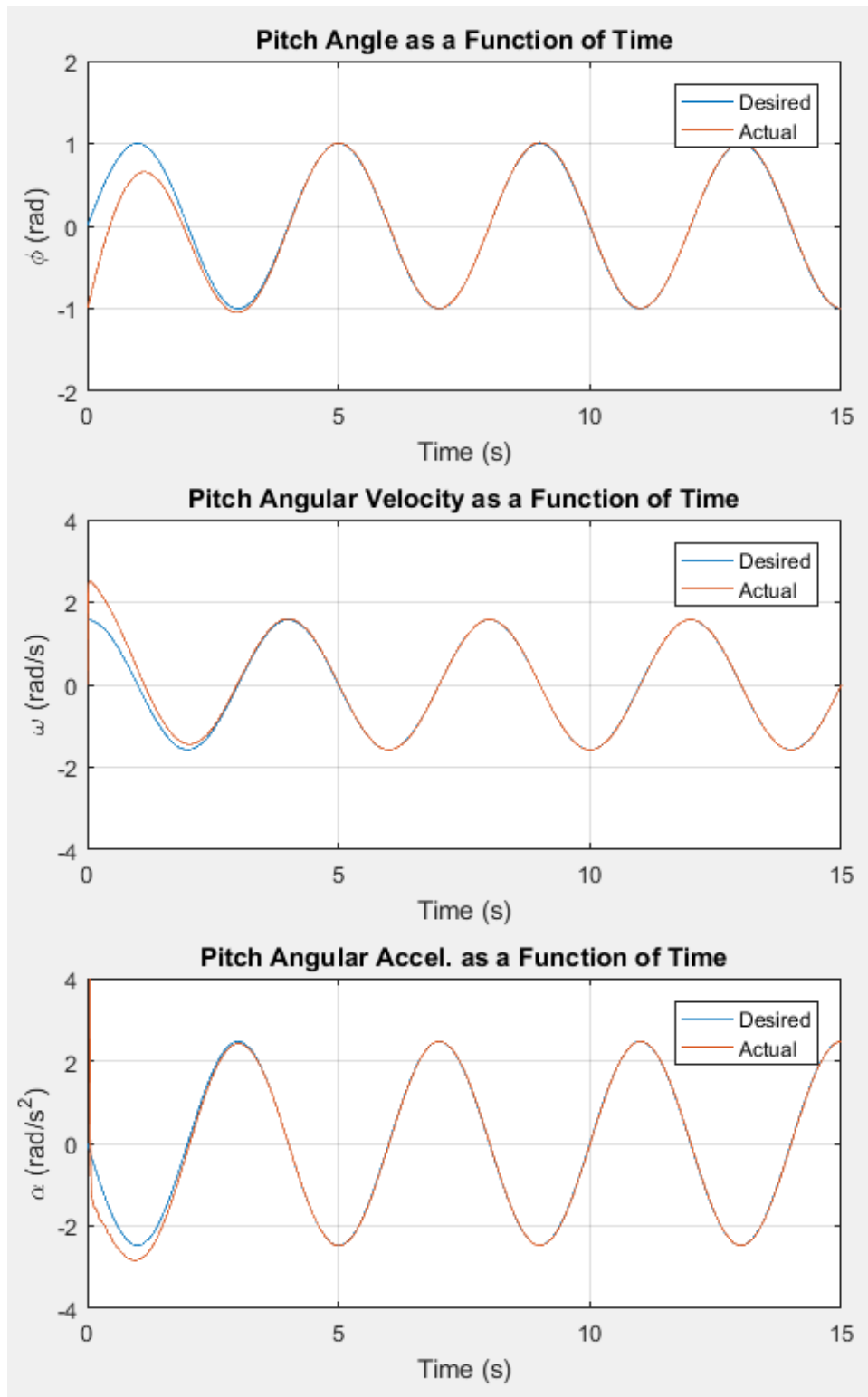


Figure 10. Closed loop simulation case 2; tracking ability.

The quadrotor is tracking the trajectory well after 3 seconds.

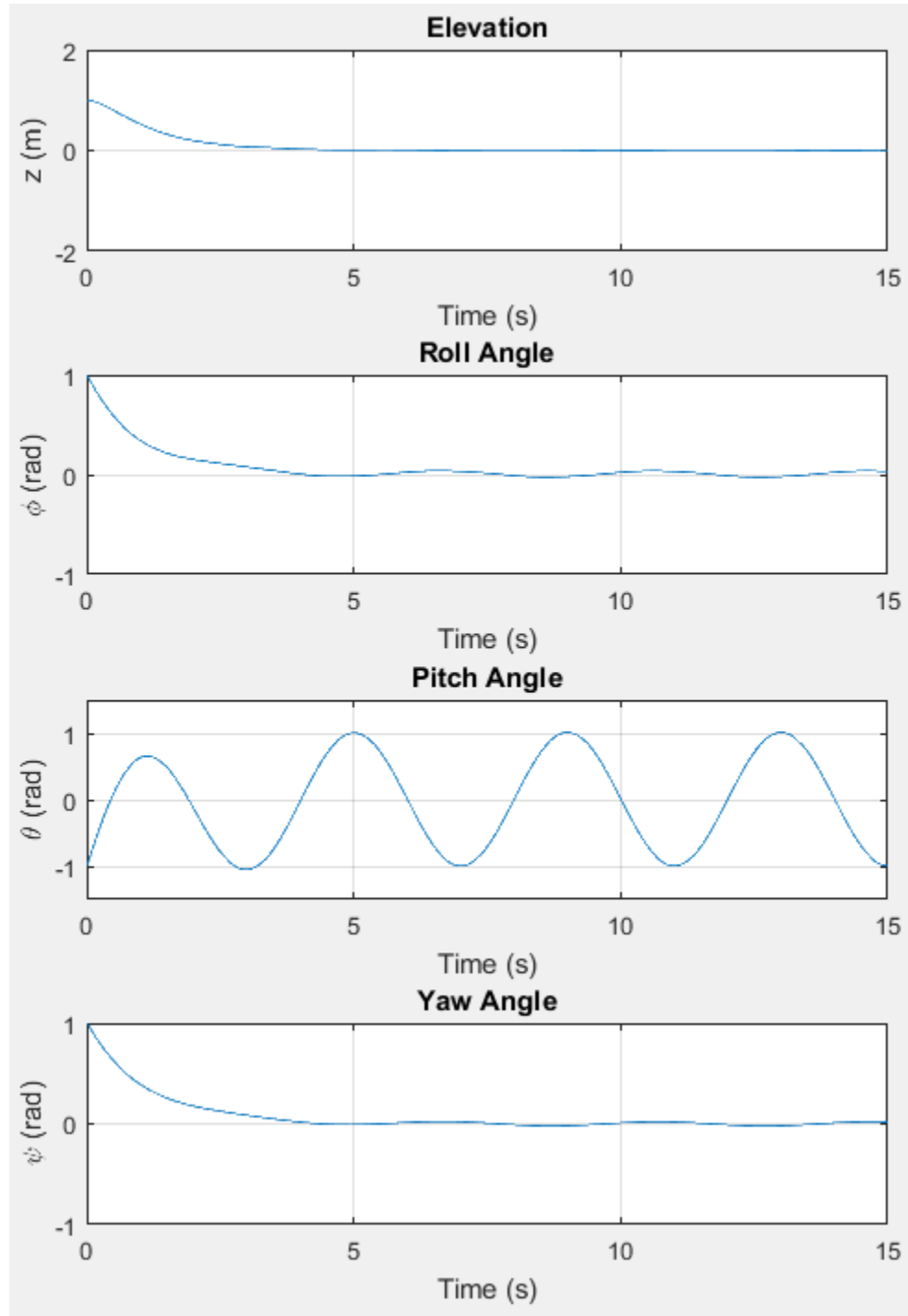


Figure 11. Closed loop simulation case 2; elevation and angles.

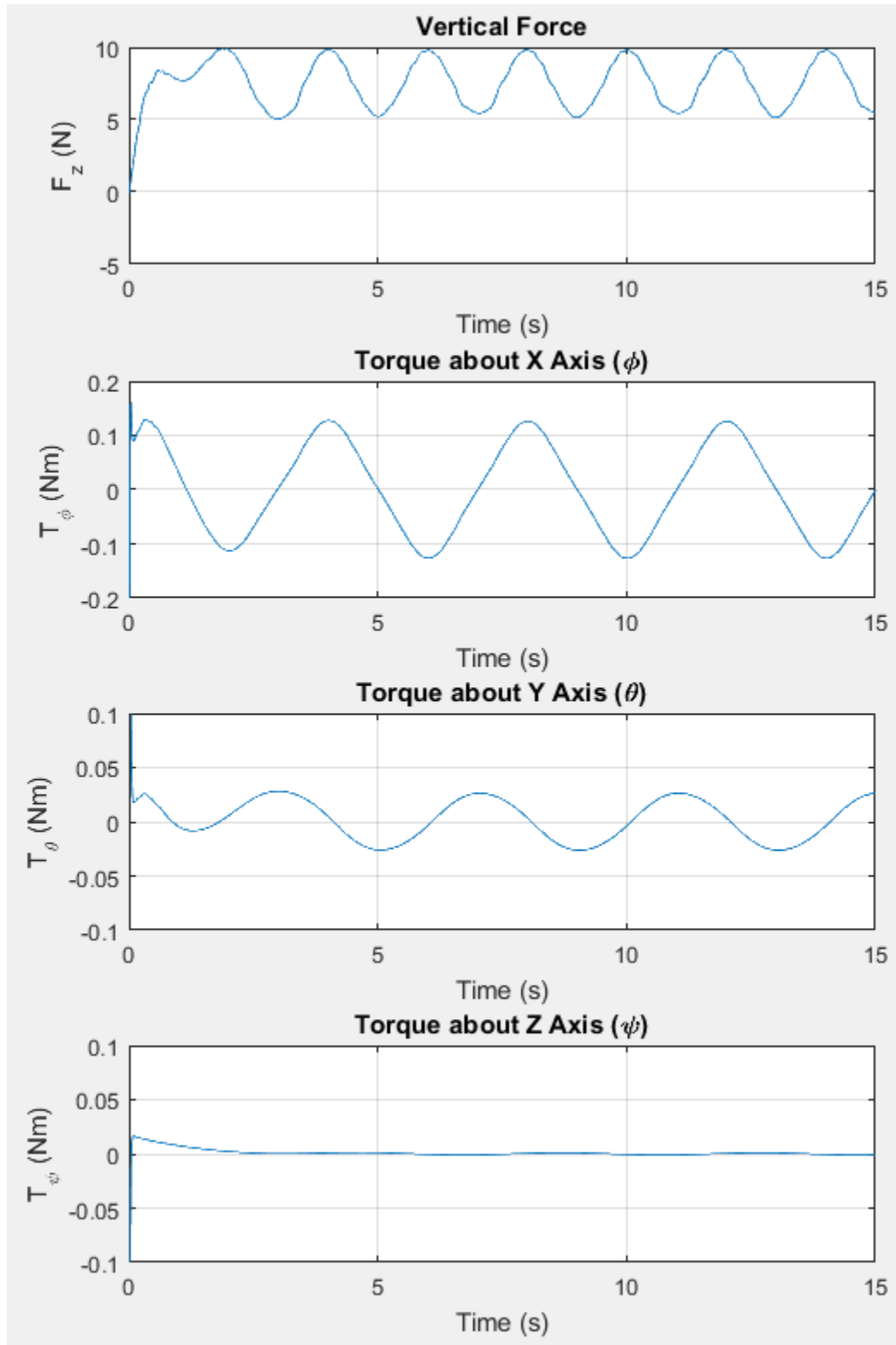


Figure 12. Closed loop simulation case 2; input vertical force and torques.

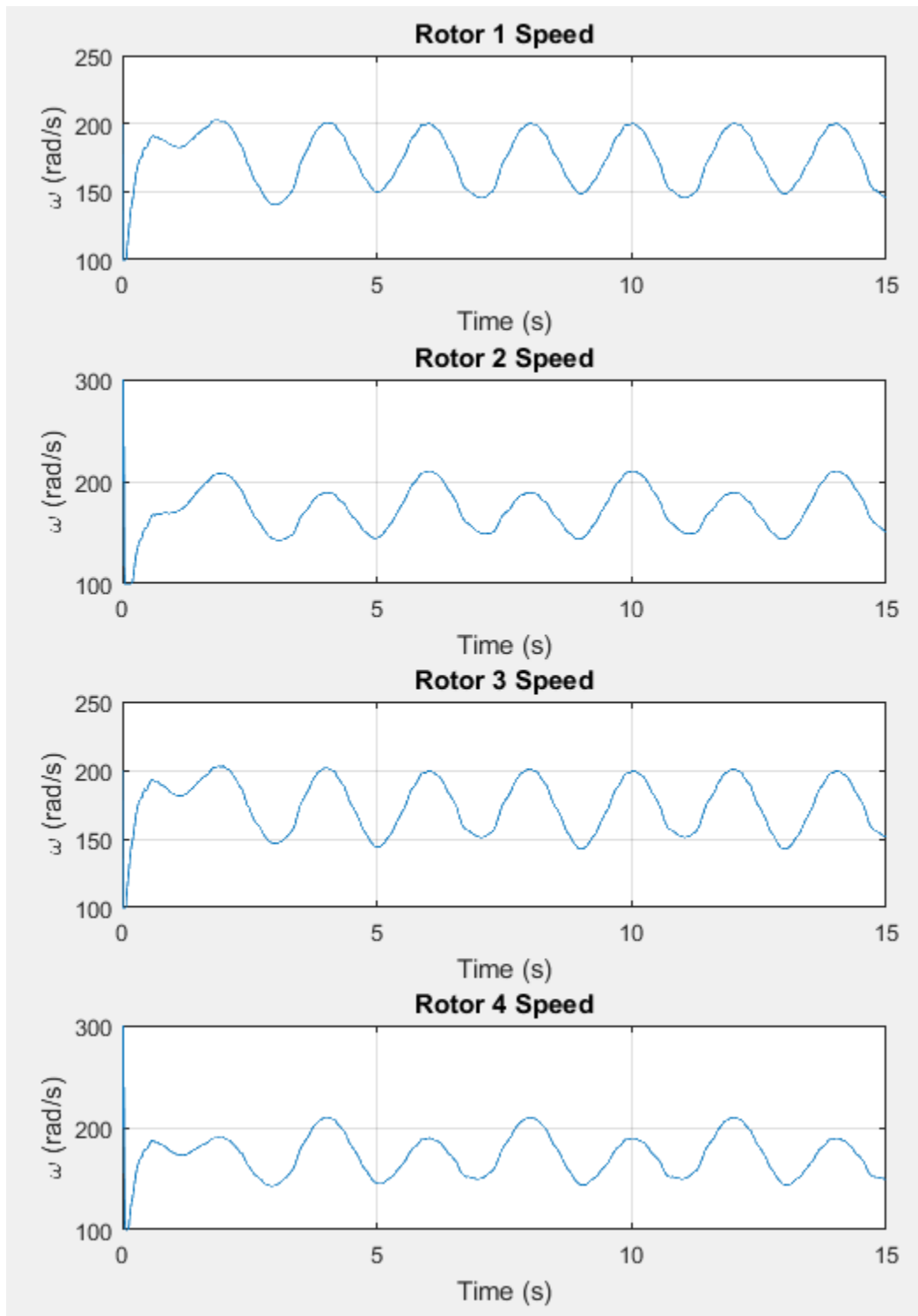


Figure 13. Closed loop simulation case 2; rotor speeds.

The rotor speeds are characterized by somewhat erratic behavior, indicative of approximation errors in the nonlinear control terms and possibly other dynamics effects.

2.4.3 Simulation Case 3

The last closed loop simulation was performed to test the quadrotor's tracking ability when subjected to a more aggressive pitch angle sinusoidal trajectory input.

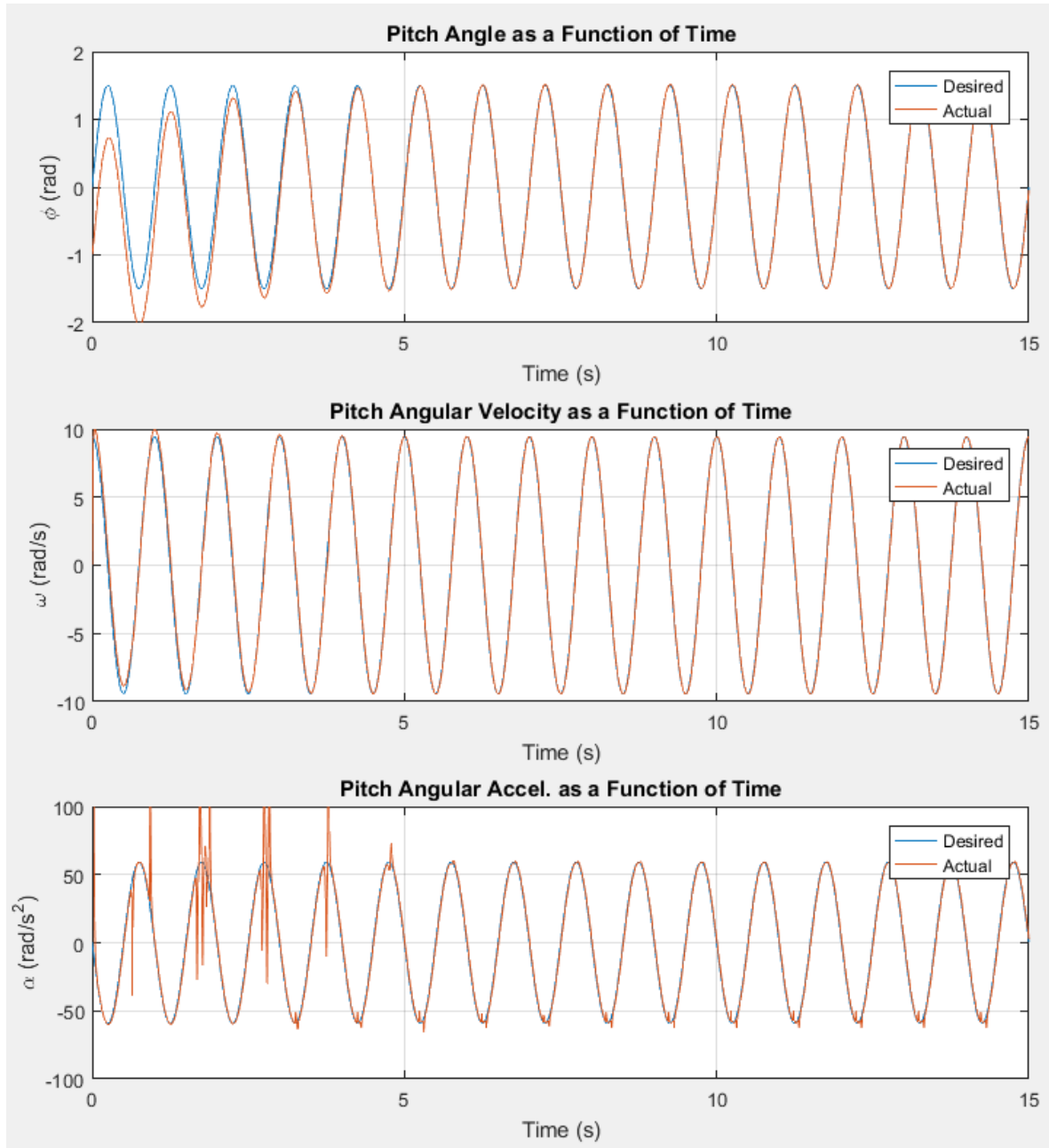


Figure 14. Closed loop simulation case 3; tracking ability.

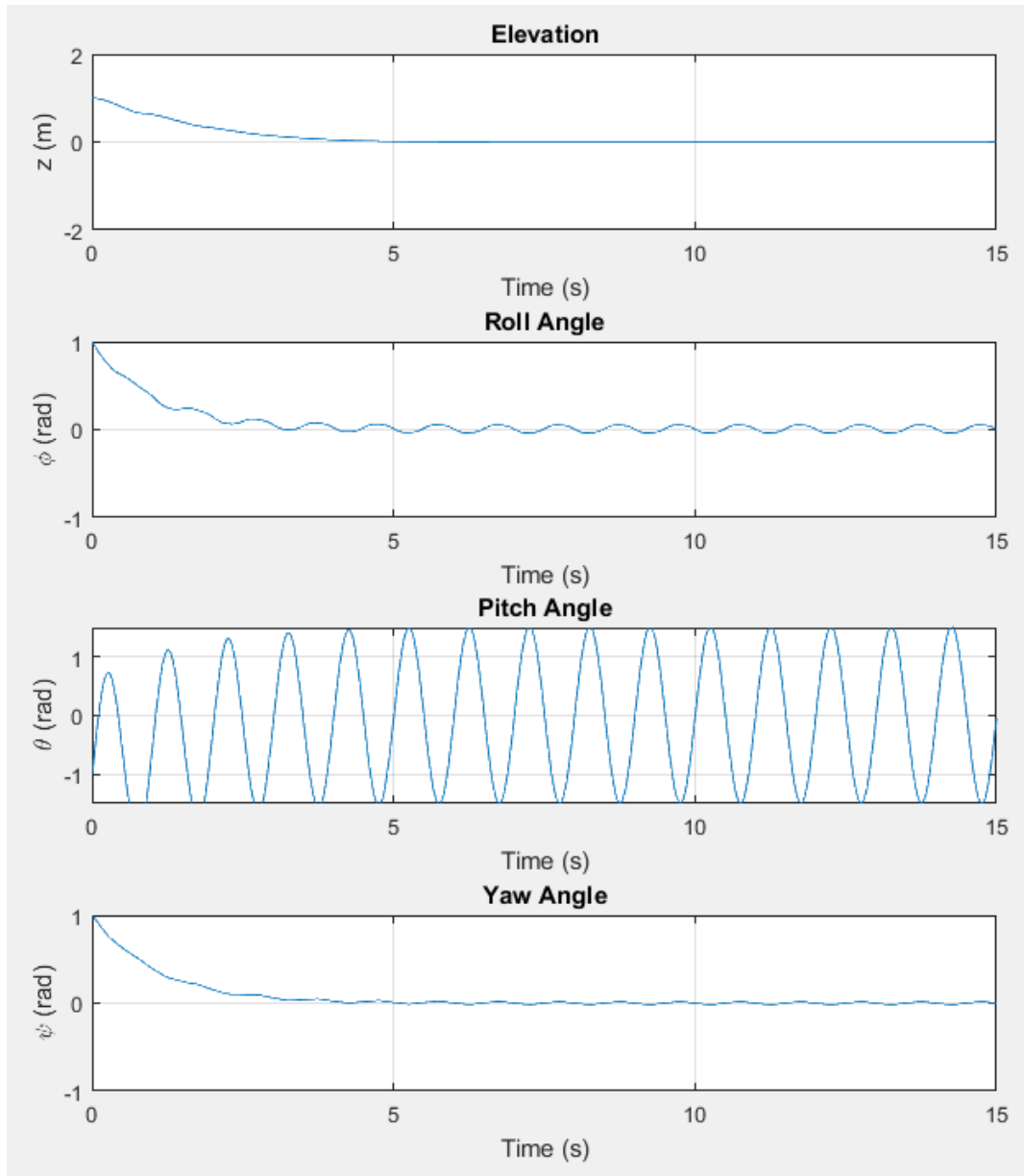


Figure 15. Closed loop simulation case 3; elevation and angles.

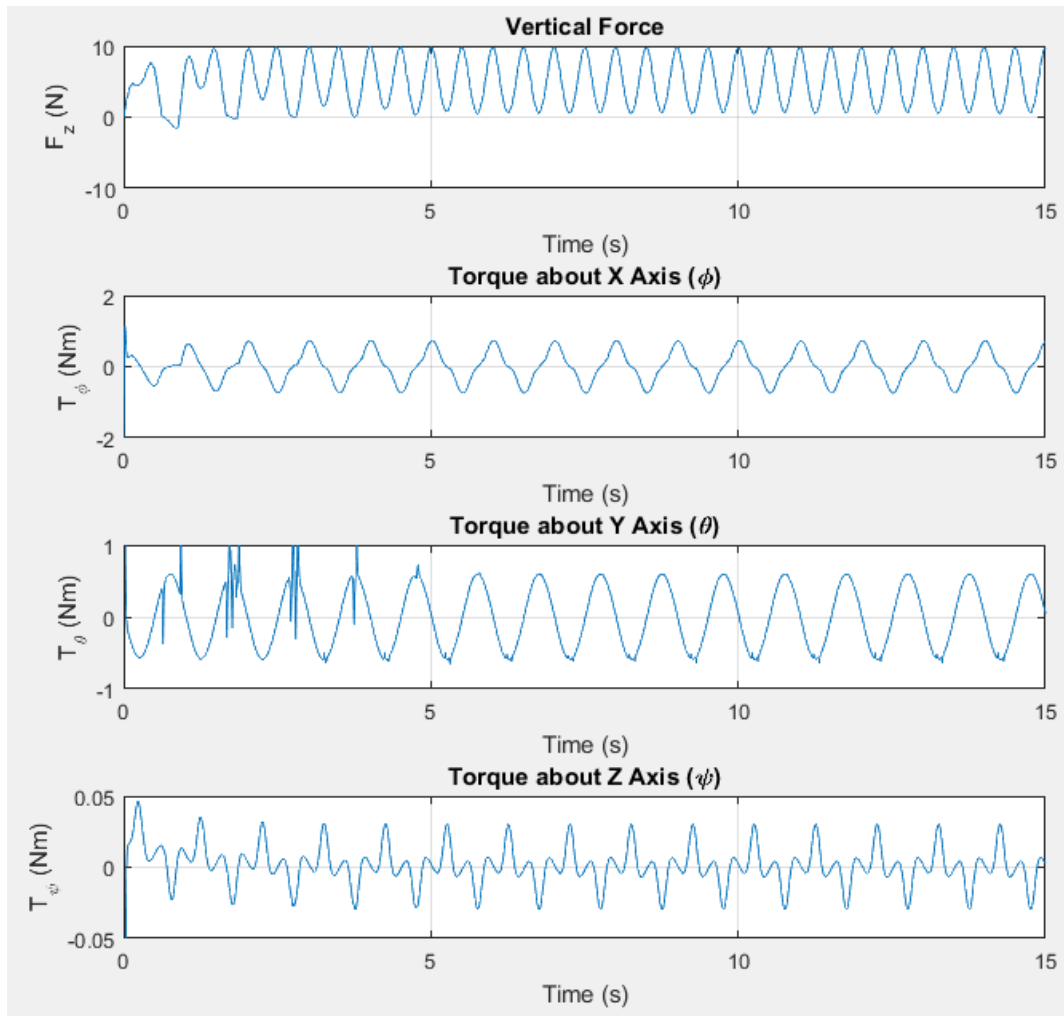


Figure 16. Closed loop simulation case 3; input vertical force and torques.

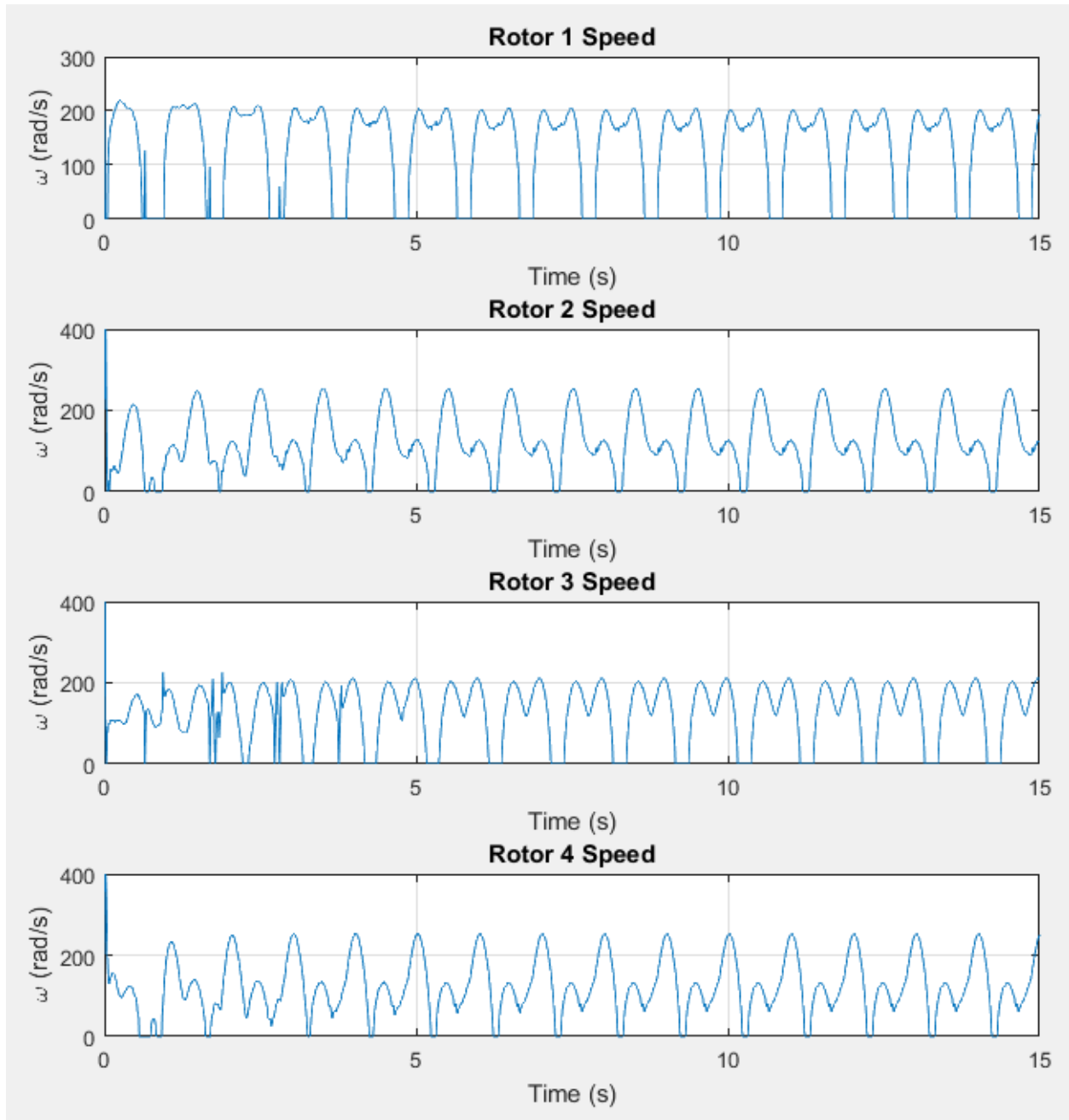


Figure 17. Closed loop simulation case 3; rotor speeds.

3 CONCLUSIONS

The objective set out for this project was achieved. An intelligent control approach was used to derive an adequate control for a quadrotor helicopter using static training. Since quadrotors have dynamics that are inherently nonlinear, the control was developed to remove nonlinearities and add linear feedback. Offline training was done using both CMAC and MLP methods. Nonlinear functions were first approximated using MLP; however, this method did not yield satisfactory results for one of the functions. Consequently, CMAC was attempted on the suspect function with significantly improved results. Simulation of a generic quadrotor was performed using the derived control with positive results for performance. Both moderate and aggressive sinusoidal trajectories were applied with reasonably accurate tracking ability and response. The more aggressive trajectory tracking performance was surprisingly solid. The analysis did not incorporate electronics and motor dynamics. A consequence of this is that simulated performance was probably more responsive than it would otherwise be.

Approximation of the nonlinear functions proved to be a challenging exercise; significant experimentation effort was required.

REFERENCES

- [1] S. Bouabdallah and R. Siegwart, "Backstepping and Sliding-mode Techniques Applied to an Indoor Micro Quadrotor," in *International Conference on Robotics and Automation*, Barcelona, Spain, 2005.
- [2] C. Nicol, C. Macnab and A. Ramiriz-Serrano, "Robust Adaptive Control of a Quadrotor Helicopter," *Elsevier Mechatronics*, 2011.

APPENDIX I

Simulink Model Diagram

